



The Ross Ice Shelf

The world's largest floating body of ice, approximately the same size as mainland France



△ THE BARRIER

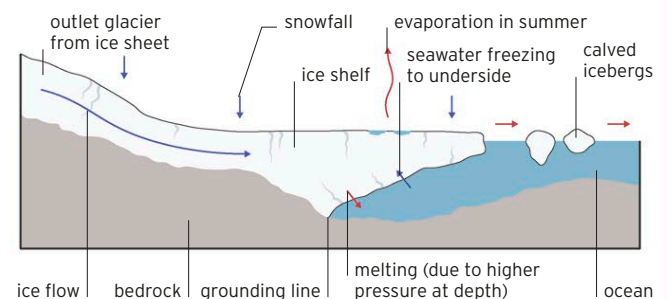
The near-vertical ice cliff at the front end of the Ross Ice Shelf is up to 165 ft (50 m) high. Underwater, it plunges down another 980 ft (300 m) or so.

The Ross Ice Shelf is a roughly triangular area of floating ice that extends up to 280 miles (450 km) from the shore of Antarctica, over the southern part of the Ross Sea. It ranges in thickness from about 1,150 ft (350 m) at its front end to around 2,460 ft (750 m) farther back, close to where its base is in contact with land. Named after Captain James Ross, who discovered it in 1841, the ice shelf was originally called the Barrier. The ice comprising the shelf flows seaward at a rate of about 2,950 ft (900 m) per year and is fed by several outlet glaciers and ice streams that flow down into it from the East Antarctic Ice Sheet (see p.306–307).

These include the Byrd, Nimrod, Beardmore, Scott, Shackleton, and Amundsen glaciers. The upper surface of the Ross Ice Shelf is an inhospitable place, subject to strong cold winds that shape it into a series of ridges and troughs called sastrugi.

ICE SHELF GAINS AND LOSSES

An ice shelf gains ice from glaciers flowing into its landward end, from water freezing to its underside, and from new snowfall. It loses ice mainly from icebergs calving (breaking off) from its front edge, and also as a result of some melting and evaporation.



◁ SHELF SHELTERERS

Some emperor penguin colonies occupy the shores of the Ross Sea a short distance from the base of the ice shelf, where the steep cliffs provide some protection against wind.